

PROJECT INITIALIZATION AND PLANNING PHASE

Date	27 June 2026
Team ID	ghousiyaunnissa@gmail.com
Project Title	SmartLender - Credit Card Approval Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform credit card issuance and approval workflows using machine learning, boosting efficiency and data-driven accuracy. It tackles manual system inefficiencies, promising faster turnaround times, reduced financial default risks, and enhanced customer satisfaction. Key features include a machine learning-based credit assessment model and real-time automated decision-making.

Project Overview

Objective	The primary objective is to streamline and automate the credit card approval process by implementing advanced machine learning classification techniques, ensuring faster, secure, and highly accurate risk assessments.
Scope	The project comprehensively analyzes historical applicant financial records, credit history, and demographics, incorporating machine learning to build a robust, dynamic, and efficient automated credit evaluation system.

Problem Statement

Description	Traditional and manual credit card application processing is highly time-consuming, prone to human error or bias, and operationally inefficient, leading to delays and inconsistent risk evaluation.
Impact	Solving these issues will result in accelerated application turnaround times, significantly minimized financial credit risks, optimized operational efficiency, and an enhanced user onboarding experience.

Proposed Solution

Approach	Employing state-of-the-art machine learning classification models to evaluate applicant profiles, predict creditworthiness, and securely output an automated approval or rejection decision.
Key Features	• Implementation of an end-to-end machine learning-based credit risk assessment model. • Real-time automated decision-making for quicker and more secure card approvals. • Continuous learning pipeline to naturally adapt to evolving financial profiles and trends.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Modern Core i5/i7/i9 or equivalent GPU
Memory	RAM specifications	8 GB / 16 GB RAM
Storage	Disk space for data, models, and logs	512 GB / 1 TB SSD
Software		
Frameworks	Python frameworks	Flask / FastAPI
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, VS Code / PyCharm
Data		
a Data	Source, size, format	Kaggle Credit Card Approval Dataset, .csv format